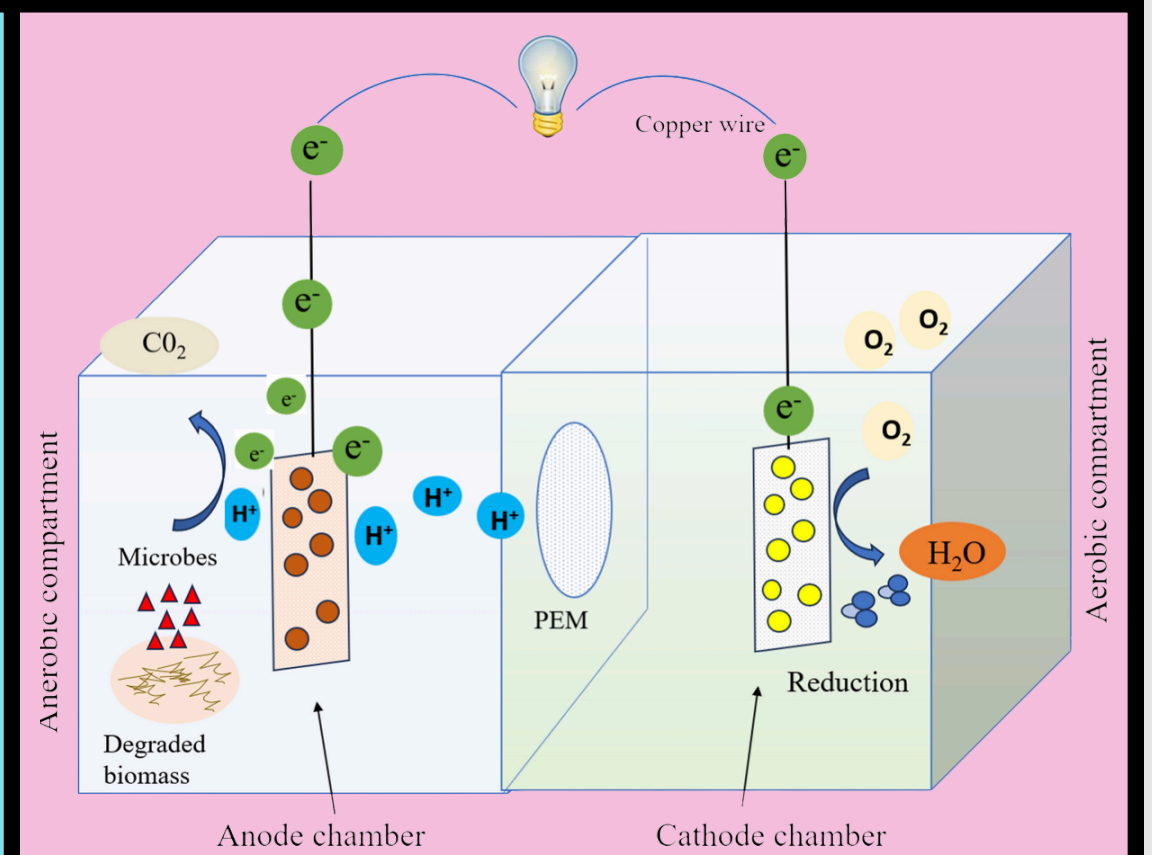


MICROBIAL FUEL CELLS

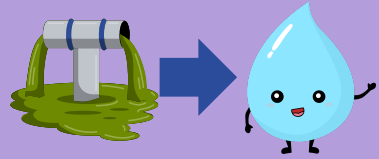



Introduction

Microbial fuel cells (MFCs) are fuel cells that **generate electric current** by using the **electrons derived from biochemical reactions** catalysed by bacteria.

There is usually a **cathode and an anode**, where electrons released in the anode by electron donors during oxidation is transferred to the cathode with electron acceptors. As most microbial cells are electrochemically inactive, mediators are used to aid electron transfer. There are **many types of MFCs**, as any organic matter could be used as fuel.

Applications



MFCs are applied in **treatment of wastewater** that is produced from industrial areas as they can **degrade organic matter**. Not only are they effective in producing cleaner water, they also

have **lower aeration requirements** (a measure of how mixed oxygen and the substrate are) compared to traditional methods like activated sludge treatment, effectively reducing the energy requirement of the system. MFCs also function as biosensors, identifying the presence of certain microbes in an environment by analysis of the produced electrical outputs.



Photo of mushroom MFC

Importance



Population growth and industrial development have increased global energy demand. This demand is largely met by fossil fuels, which contribute to global warming. This highlights the importance of **renewable energy alternatives**.

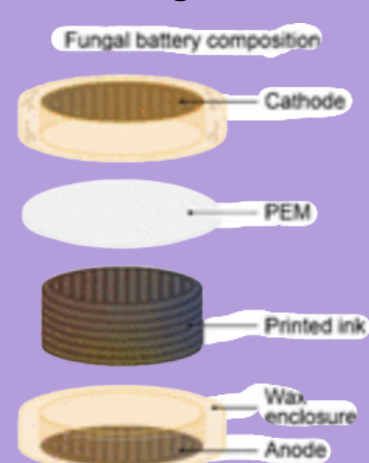
MFCs rely on microbial processes, which means their **carbon footprint is significantly smaller**. They also **reduce methane emissions** by controlling how organic matter breaks down as they use it to generate electricity release instead of releasing methane into the atmosphere.

They are also **accessible to less developed areas**. This is because they generate electricity internally and they thus require minimal external power.

Case study

3D-printed MFC with fungal cells using yeast and white rot fungi!

For a single MFC, a maximum power of $12.54 \mu\text{W}/\text{cm}^2$ was obtained at a current density of $49.2 \mu\text{A}/\text{cm}^2$ with a $22 \text{ k}\Omega$ load attached to the MFC.



This is a greener alternative for low power devices.

Conclusion

However, microbial fuel cells are not without their limitations. They **generate relatively low power** compared to conventional energy sources. Some of the materials required to produce microbial fuel cells are also **expensive**.

Despite all this, microbial cells are still promising for renewable energy research.



Super cool links!

https://en.wikipedia.org/wiki/Microbial_fuel_cell
<https://pubs.acs.org/doi/10.1021/acssuschemeng.4c05494>
<https://pmc.ncbi.nlm.nih.gov/articles/PMC10223362/>
<https://www.mdpi.com/2673-6284/11/4/44>